

Design and implementation of an autonomous mobile robot based on computer vision and QR recognition for controlled logistics environments

Diseño e implementación de un robot móvil autónomo basado en visión artificial y reconocimiento de códigos QR para entornos logísticos controlados

Vanesa Galeano¹, Julián García², José Hurtado³, Sofía Espitia⁴

ABSTRACT

This paper presents the design and implementation of an autonomous mobile robot named ATLAS for navigation and material transport tasks in controlled logistics environments. The system employs a distributed architecture based on Raspberry Pi 5 and ESP32 for computer vision processing and motion control. The robot integrates Ackermann steering, JGY370 motors, and an IMX219 camera for path tracking and QR code recognition using algorithms developed in Python and OpenCV. Experimental validation was performed on a test track with loading and unloading stations, evaluating line following, QR detection, and automatic stopping tasks. Experimental results validate the functional integration of embedded systems, computer vision, and autonomous navigation in a robotic platform for controlled logistics applications.

Keywords: autonomous mobile robot, computer vision, autonomous navigation, embedded systems, logistics automation.

RESUMEN

Se presenta el diseño e implementación de un robot móvil autónomo denominado ATLAS para tareas de navegación y transporte de materiales en entornos logísticos controlados. El sistema emplea una arquitectura distribuida basada en Raspberry Pi 5 y ESP32 para procesamiento de visión artificial y control de movimiento. El robot integra dirección tipo Ackermann, motores JGY370 y una cámara IMX219 para seguimiento de trayectoria y reconocimiento de códigos QR mediante algoritmos desarrollados en Python y OpenCV. La validación experimental se realizó sobre una pista de pruebas con estaciones de carga y descarga, evaluando seguimiento de línea, detección de códigos QR y detención automática. Los resultados experimentales validan la integración funcional de sistemas embebidos, visión artificial y navegación autónoma en una plataforma robótica orientada a aplicaciones logísticas controladas.

Palabras clave: robot móvil autónomo, visión artificial, navegación autónoma, sistemas embebidos, automatización logística.

¹Undergraduate Student in Mechatronics Engineering, Universidad Tecnológica de Pereira, Colombia. Affiliation: Mechatronics Engineering Program, Universidad Tecnológica de Pereira, Colombia. Email: vanesa.galeano@utp.edu.co

²Undergraduate Student in Mechatronics Engineering, Universidad Tecnológica de Pereira, Colombia. Affiliation: Mechatronics Engineering Program, Universidad Tecnológica de Pereira, Colombia. Email: julandgarcia@utp.edu.co

³Undergraduate Student in Mechatronics Engineering, Universidad Tecnológica de Pereira, Colombia. Affiliation: Mechatronics Engineering Program, Universidad Tecnológica de Pereira, Colombia. Email: j.hurtado1@utp.edu.co

⁴Undergraduate Student in Mechatronics Engineering, Universidad Tecnológica de Pereira, Colombia. Affiliation: Mechatronics Engineering Program, Universidad Tecnológica de Pereira, Colombia. Email: sofia.espitia@utp.edu.co



Introduction

Autonomous mobile robots (AMRs) have gained significant relevance in logistics and industrial automation due to their ability to optimize material transport, inventory handling, and internal distribution processes in structured environments [1]. The integration of embedded systems, computer vision, and autonomous navigation strategies has enabled the development of flexible robotic platforms aligned with Industry 4.0 requirements and smart warehouse applications [2]. Recent advances in autonomous robotics have focused on improving navigation accuracy, reducing operational costs, and increasing system adaptability in controlled logistics scenarios.

Computer vision has become one of the most widely implemented perception strategies in AMR systems because it enables real-time environment analysis using low-cost sensors and embedded processing platforms [3]. Vision-based navigation techniques allow robots to perform tasks such as line following, visual marker recognition, and route identification without requiring complex external infrastructure [4]. In logistics applications, QR codes have been adopted as reliable visual references for localization and task execution due to their low computational requirements, ease of implementation, and capacity to store navigation-related information [5].

Despite recent advances in autonomous navigation, several AMR implementations still depend on high-cost sensors, centralized processing architectures, or computationally intensive localization methods that limit their deployment in small-scale or controlled environments [1], [3]. Additionally, real-time integration between perception and locomotion systems remains a challenge in embedded robotic platforms, particularly when low-cost hardware architectures are employed. These limitations motivate the development of distributed robotic systems capable of separating perception and control tasks while maintaining stable autonomous operation.

This work presents the design and implementation of an autonomous mobile robot named ATLAS for controlled logistics navigation and material transport tasks. The proposed system employs a distributed architecture based on Raspberry Pi 5 and ESP32, where the Raspberry Pi performs computer vision processing and navigation decision-making, while the ESP32 executes low-level locomotion control through bidirectional serial communication. The robotic platform integrates Ackermann steering, JGY370 motors, an IMX219 camera, and computer vision algorithms developed in Python and OpenCV for line following and QR code recognition.

Experimental validation was conducted on a structured test track containing loading and unloading stations. The evaluation considered line-following performance, QR code detection, and automatic stopping tasks under autonomous operation conditions. The main contributions of this work include the

implementation of a distributed embedded architecture for autonomous navigation, the integration of computer vision and QR-based guidance in a mobile robotic platform, and the experimental validation of autonomous logistics tasks in a controlled environment.

Methodology

The development of the ATLAS autonomous mobile robot was carried out using a Project-Based Learning (PBL) approach combined with iterative prototyping and experimental validation stages. The methodology integrated mechanical design, embedded system implementation, computer vision processing, communication architecture development, and functional testing in a controlled logistics environment.

Table 1. Hardware specifications

COMPONENT	SPECIFICATION
Main processor	Raspberry Pi 5
Microcontroller	ESP32
Camera	IMX219
Motor driver	TB6612FNG
Traction motors	JGY370 6V
Steering system	Ackermann
Programming languages	Python, C++

Source: Authors

A. System Architecture

The proposed robotic platform employed a distributed architecture based on a Raspberry Pi 5 and an ESP32 microcontroller. The Raspberry Pi was responsible for high-level tasks, including computer vision processing, QR code recognition, trajectory analysis, and navigation decision-making. The ESP32 executed low-level control tasks related to motor actuation and locomotion control through PWM signals. Both devices communicated through bidirectional serial communication to coordinate navigation commands and system responses.

This distributed approach reduced computational load on the control system and improved real-time response during autonomous operation. Fig. 1 illustrates the general architecture

of the proposed system.

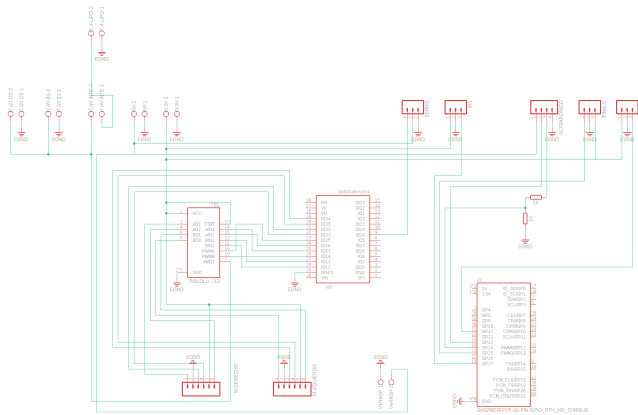


Figure 1. Distributed architecture of the ATLAS autonomous mobile robot.

B. Mechanical Design

The robot chassis was designed to support autonomous navigation and material transport tasks in structured environments. The mechanical configuration incorporated an Ackermann steering mechanism to improve trajectory stability and directional control during displacement. The traction system used JGY370 6V DC motors coupled to the rear axle, while the front axle implemented the steering mechanism.

The structural components were manufactured using 3D printing techniques to facilitate rapid prototyping and modular assembly. The final platform was designed to support loads up to 1 kg while maintaining mechanical stability during navigation tests.

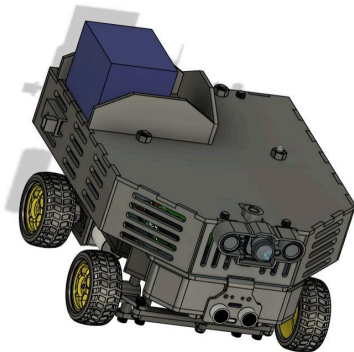


Figure 2. Mechanical structure of the ATLAS mobile robot platform.
Source: Authors

C. Embedded Control System

The embedded control system was implemented using an ESP32 microcontroller and a TB6612FNG motor driver. The ESP32 generated PWM control signals for motor actuation and managed the execution of movement commands received from the Raspberry Pi.

The control logic included forward motion, steering adjustment, stopping routines, and speed regulation according to the

navigation conditions detected by the vision system. Serial communication was implemented to ensure synchronized operation between perception and locomotion modules.

D. Computer Vision and Navigation

The perception system employed an IMX219 camera connected to the Raspberry Pi 5. Computer vision algorithms were developed in Python using the OpenCV library for line-following and QR code recognition tasks.

The navigation strategy was based on trajectory detection through image processing techniques that identified the reference path on the test track. Additionally, QR codes positioned at specific locations were used as visual markers to trigger predefined actions such as stopping at loading stations and resuming navigation.

Image acquisition and processing were executed in real time to maintain continuous autonomous operation during the experimental tests.

E. Experimental Validation

Experimental validation was conducted on a controlled test track designed with navigation routes and loading/unloading stations. Three functional scenarios were evaluated:

- Line-following navigation.
- QR code recognition.
- Automatic stopping at loading stations.

The validation process analyzed the robot's autonomous response, trajectory tracking capability, communication stability, and integration between embedded control and computer vision systems. The experiments verified the functional operation of the robotic platform under structured navigation conditions.

Table 2. Experimental validation scenarios

TEST	DESCRIPTION	VALIDATION OBJECTIVE
Test 1	Line following	Trajectory tracking
Test 2	QR recognition	Visual marker detection
Test 3	Automatic stopping	Autonomous station handling

Source: Authors

Results

The experimental validation of the ATLAS autonomous mobile robot was performed on a structured test track designed to evaluate navigation, perception, and autonomous response under controlled logistics conditions. The experiments focused on line-following performance, QR code recognition, and automatic stopping at loading and unloading stations.

A. Line-Following Performance

The robot successfully executed autonomous trajectory tracking using computer vision algorithms implemented in Python and OpenCV. The IMX219 camera continuously captured images of

the navigation path, allowing the Raspberry Pi 5 to process visual information in real time and generate steering corrections through the distributed control architecture.

During the tests, the Ackermann steering mechanism provided stable directional control while the ESP32 maintained synchronized motor actuation through PWM signals. The robot maintained continuous movement along the predefined route without requiring external intervention. Fig. 3 presents the line-following validation scenario.

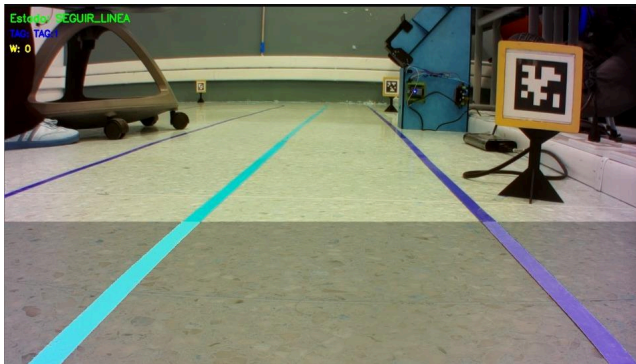


Figure 3. Computer vision processing for line following
Source: Authors

B. QR Code Recognition

QR code detection was evaluated using visual markers positioned at specific locations along the test track. The implemented image-processing algorithms successfully identified QR codes corresponding to predefined navigation actions.

The recognition process was executed in real time using the OpenCV library, enabling the robot to interpret navigation instructions and trigger operational events during autonomous displacement. The distributed processing architecture allowed stable communication between the perception and locomotion modules during QR detection tasks. Fig. 4 illustrates the QR recognition process implemented in the experimental environment.

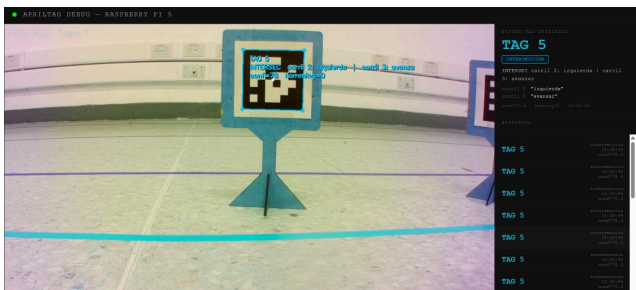


Figure 4. Computer vision processing for QR code detection.
Source: Authors

C. Automatic Stopping at Loading Stations

The robot demonstrated autonomous stopping capability at loading and unloading stations after detecting predefined QR

markers. Once a station identifier was recognized, the Raspberry Pi transmitted stopping commands to the ESP32 through bidirectional serial communication.

The embedded control system executed controlled motor deceleration and temporary stopping routines before resuming navigation. The experiments verified stable communication between the processing and control modules during station-handling operations. Fig. 5 presents the experimental validation of the automatic stopping process.

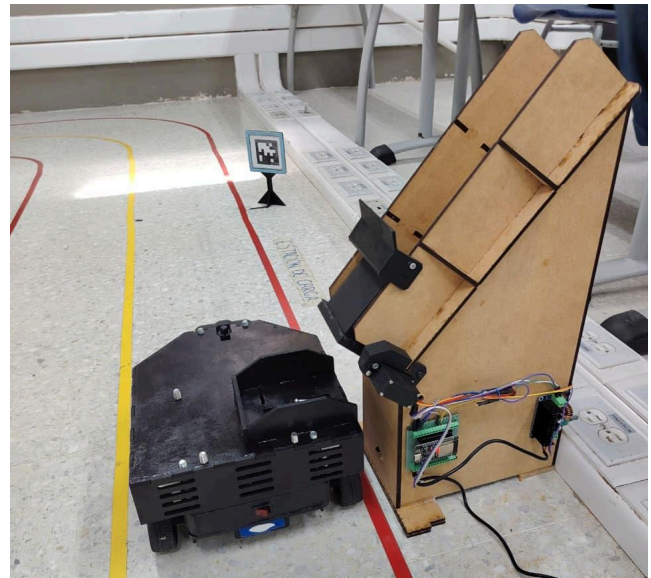


Figure 5. Experimental validation environment with loading and unloading stations.
Source: Authors

D. System Integration and Autonomous Operation

The experimental results validated the functional integration of computer vision, embedded control, distributed communication, and autonomous navigation within the robotic platform. The Raspberry Pi 5 and ESP32 architecture enabled the separation of perception and locomotion tasks while maintaining stable real-time operation during the navigation scenarios.

The implemented system achieved autonomous navigation under structured environmental conditions, demonstrating the feasibility of integrating low-cost embedded hardware and computer vision techniques for controlled logistics applications.

Discussion

The experimental results demonstrated that the proposed distributed architecture enabled stable integration between computer vision, embedded control, and autonomous navigation tasks in a controlled logistics environment. The separation of high-level perception processes and low-level locomotion control between the Raspberry Pi 5 and ESP32 reduced computational overload and improved real-time system response during autonomous operation. Similar distributed approaches have been

reported in autonomous robotic systems to improve processing efficiency and operational stability in embedded platforms.

The implementation of computer vision algorithms using Python and OpenCV allowed reliable trajectory tracking and QR code recognition under structured environmental conditions. Compared with navigation strategies based on complex localization systems or high-cost sensors, the proposed approach provides a low-complexity alternative for autonomous navigation in controlled scenarios [1], [3]. The use of QR codes as visual markers simplified route identification and station handling processes while maintaining stable detection during the experiments.

The Ackermann steering configuration contributed to improved directional stability during trajectory tracking tasks. The experimental tests showed that the robot maintained continuous navigation without requiring external intervention, demonstrating functional coordination between perception and locomotion modules. Additionally, the serial communication implemented between the Raspberry Pi and ESP32 enabled synchronized execution of navigation commands and motor control actions during autonomous operation.

Although the implemented system achieved stable performance under controlled conditions, some limitations were identified during the validation process. Variations in illumination conditions affected image acquisition quality and could influence trajectory detection accuracy in environments with non-uniform lighting. Similarly, the navigation strategy depended on predefined visual references and structured paths, limiting system adaptability in dynamic or unstructured environments.

Future work may include the integration of simultaneous localization and mapping (SLAM) techniques, obstacle avoidance strategies, and deep learning-based perception methods to improve navigation robustness and environmental adaptability. Additionally, the incorporation of wireless communication and cloud-based monitoring systems could extend the applicability of the robotic platform to larger logistics environments and collaborative automation scenarios.

Conclusions

This work presented the design and implementation of an autonomous mobile robot named ATLAS for controlled logistics navigation and material transport tasks. The proposed system employed a distributed architecture based on Raspberry Pi 5 and ESP32 to separate computer vision processing and locomotion control tasks, enabling stable autonomous operation during experimental validation.

The implemented computer vision system, developed in Python and OpenCV using an IMX219 camera, successfully performed line-following and QR code recognition tasks in real time. Additionally, the integration of Ackermann steering, embedded

control, and bidirectional serial communication allowed coordinated navigation and automatic stopping at loading and unloading stations.

The experimental results validated the functional integration of embedded systems, computer vision, and autonomous navigation in a robotic platform operating under structured environmental conditions. The proposed architecture demonstrated the feasibility of using low-cost embedded hardware for autonomous logistics applications in controlled scenarios.

Future work may include the incorporation of obstacle avoidance strategies, SLAM-based localization methods, and advanced perception algorithms to improve system adaptability in dynamic environments. Furthermore, wireless monitoring and multi-robot coordination strategies could extend the applicability of the proposed platform in larger automation systems.

Acknowledgements

The authors would like to thank the Mechatronics Engineering Program and the Universidad Tecnológica de Pereira for providing the academic guidance, laboratory resources, and technical support required for the development and experimental validation of the ATLAS autonomous mobile robot platform.

Author contributions

Sofía Espitia contributed mainly to system design, Vanesa Galeano to programming and computer vision tasks, Julián García to the electrical and electronic integration, and Jose Hurtado to the mechanical development of the platform. All authors collaboratively participated in system integration, experimental validation, documentation, and manuscript preparation through a cooperative development process. All authors reviewed and approved the final version of the manuscript.

Conflicts of interest

The authors declare no conflicts of interest related to this work.

References

- [1] K. Ellithy, M. Ahmed, and A. Hassan, "AGV and Industry 4.0 in warehouses: A comprehensive analysis of existing literature and an innovative framework for flexible automation," *The International Journal of Advanced Manufacturing Technology*, vol. 132, pp. 4211–4234, 2024. <https://doi.org/10.1007/s00170-024-14127-0>
- [2] M. A. El-Sayed, H. Ibrahim, and S. Abdelrahman, "AI-powered warehouse management: Autonomous mobile robots and advanced optimization with NFC, MILP, and swarm algorithms," in *Proc. IEEE International Conference on Industrial Technology*, 2024. [Online]. Available: <https://ieeexplore.ieee.org/document/11004979/>
- [3] Y. Zhang, H. Wang, and Z. Liu, "Vision-based autonomous navigation for mobile robots using embedded computer vision systems," *Sensors*, vol. 23, no. 14, pp. 1–18, 2023.

<https://www.sciencedirect.com/journal/robotics-and-autonomous-systems?utm>

- [4] G. Bradski, "The OpenCV library," Dr. Dobb's Journal of Software Tools, vol. 25, no. 11, pp. 120–125, 2000.
<https://ptacts.uspto.gov/ptacts/public-informations/petitions/1558289/download-documents?artifactId=8E9mKCVRLtZAlHYKaMMeW0Y7vRNLW2jO3lqlxClvFTojVG2bGeGwSDI>
- [5] R. Kumar, S. Patel, and A. Singh, "Mobile robotic manipulator based autonomous warehouse operations," in Proc. IEEE International Conference on Automation Science and Engineering, 2023. [Online]. Available: <https://ieeexplore.ieee.org/document/10374743>
- [6] V. K. Yarlagadda, "Cutting-edge developments in robotics for smart warehousing and logistics optimization," Robotics Xplore: USA Automation Digest, vol. 5, no. 2, pp. 15–24, 2024.
<https://roboticsxplore.com/index.php/rx/article/view/7?utm>
- [7] A. Balachandran, P. Kumar, and R. Nair, "Autonomous navigation of an AMR using deep reinforcement learning in warehouse environment," in Proc. IEEE International Conference on Intelligent Robots and Systems, 2022. [Online]. Available: <https://ieeexplore.ieee.org/document/9971804>
- [8] S. Espitia, V. Galeano, J. García, and J. Hurtado, "ATLAS autonomous mobile robot: Experimental validation platform," unpublished project manuscript, Universidad Tecnológica de Pereira, Pereira, Colombia, 2025.
https://docs.google.com/document/d/1REdrtNgITMxJ_7I2HikQf-MAyTd7sibm/edit?usp=sharing&oid=106682866766492717273&tpof=true&sd=true
- [9] "Manual de usuario: ATLAS robot autónomo — tres casos de prueba," Universidad Tecnológica de Pereira, Facultad de Tecnología, Programa de Ingeniería Mecatrónica, Pereira, Colombia, 2025.
https://drive.google.com/file/d/1QRH3H6fQlt_Ddnmmmtwcvb6_MVRJJV2W/view?usp=sharing
- [10] "Acta de constitución del proyecto robot autónomo ATLAS," Universidad Tecnológica de Pereira, Facultad de Tecnología, Programa de Ingeniería Mecatrónica, Pereira, Colombia, 2025.
<https://docs.google.com/document/d/1pEP2whKF3BkKWY76yrVcwxnDQFxJtWQt59tookOMmHM/edit?usp=sharing>
- [11] R. Siegwart, I. R. Nourbakhsh, and D. Scaramuzza, Introduction to Autonomous Mobile Robots, 2nd ed. Cambridge, MA, USA: MIT Press, 2019.
<https://mitpress.mit.edu/9780262015356/introduction-to-autonomous-mobile-robots/>
- [12] B. Siciliano and O. Khatib, Springer Handbook of Robotics. Cham, Switzerland: Springer, 2016.
<https://doi.org/10.1007/978-3-319-32552-1>